In [ ]:
```
#Advanced EDA for iris dataset
```

In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
iris = pd.read_csv(r'C:\Users\SID Balurkar\Desktop\Full Stack Data Science\Classwork\Nov 2025\26th, 27th- Advanced Ed
iris
```

Out[2]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [15]:
```
import seaborn as sns
import matplotlib.pyplot as plt

import warnings
warnings.filterwarnings('ignore')
```

In [4]:
```
iris.head()
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

In [6]:
```
iris.drop('Id', axis = 1, inplace = True)
```

In [7]:
```
iris.head()
```

Out[7]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

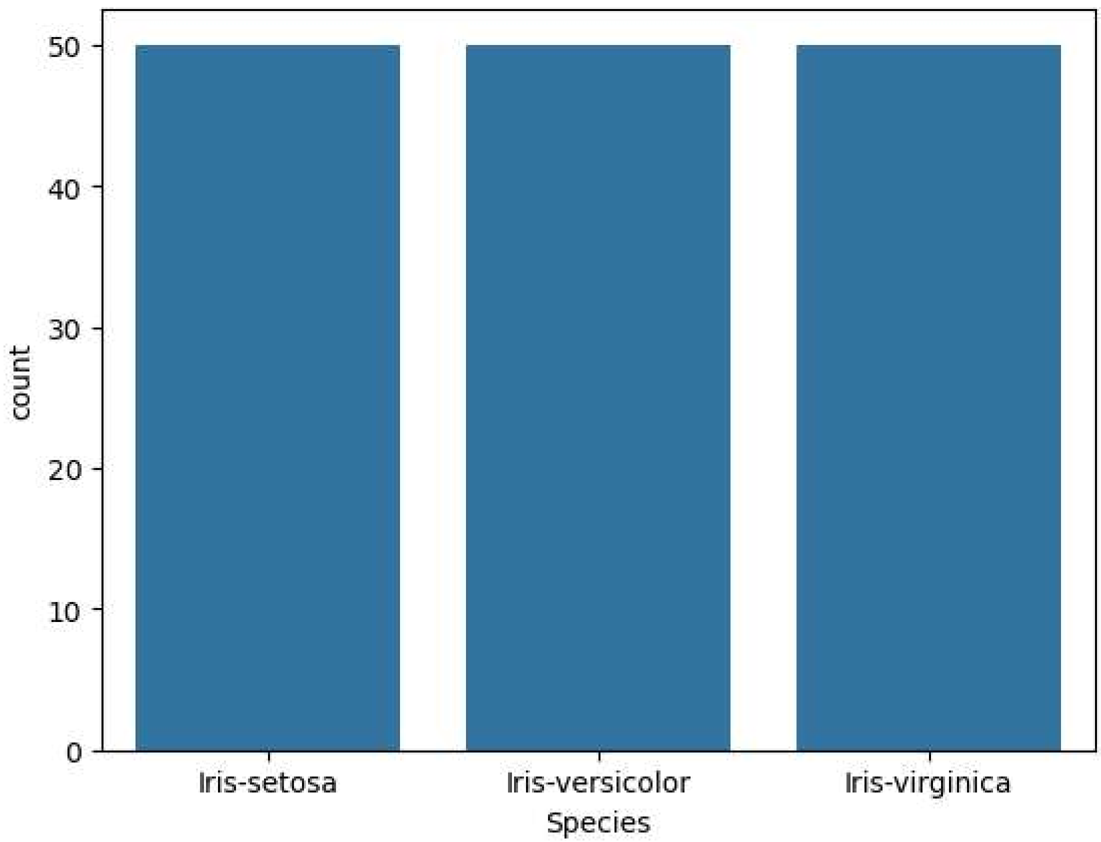
In [8]:
```
iris.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   SepalLengthCm  150 non-null    float64
 1   SepalWidthCm   150 non-null    float64
 2   PetalLengthCm  150 non-null    float64
 3   PetalWidthCm   150 non-null    float64
 4   Species        150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [9]:
```
iris['Species'].value_counts()
```

Out[9]:
```
Species
Iris-setosa        50
Iris-versicolor    50
Iris-virginica     50
Name: count, dtype: int64
```

In [20]:
```
sns.countplot(x = 'Species',data=iris)
plt.show()
```

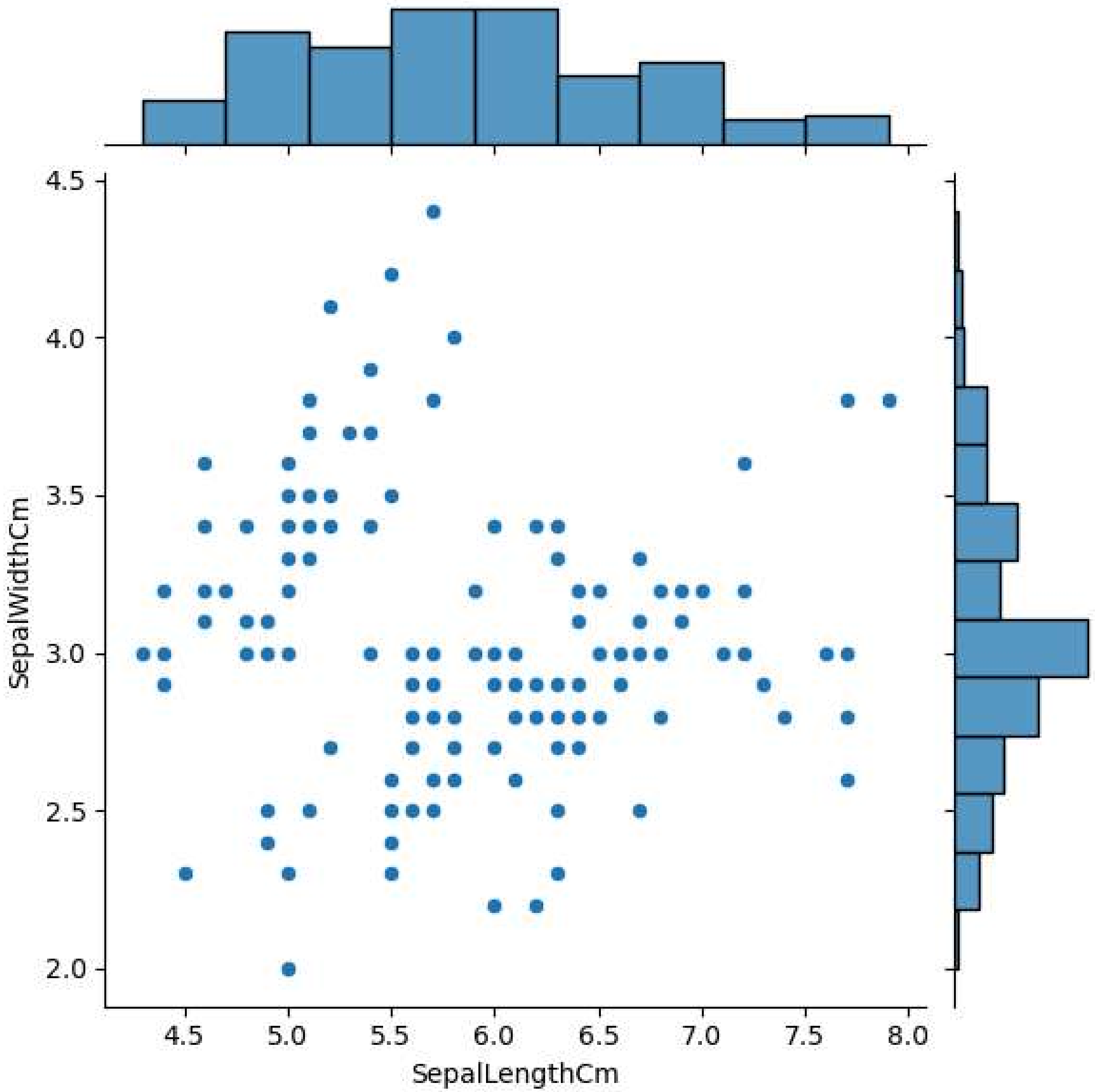


In [21]:
```
iris.head()
```

Out[21]:

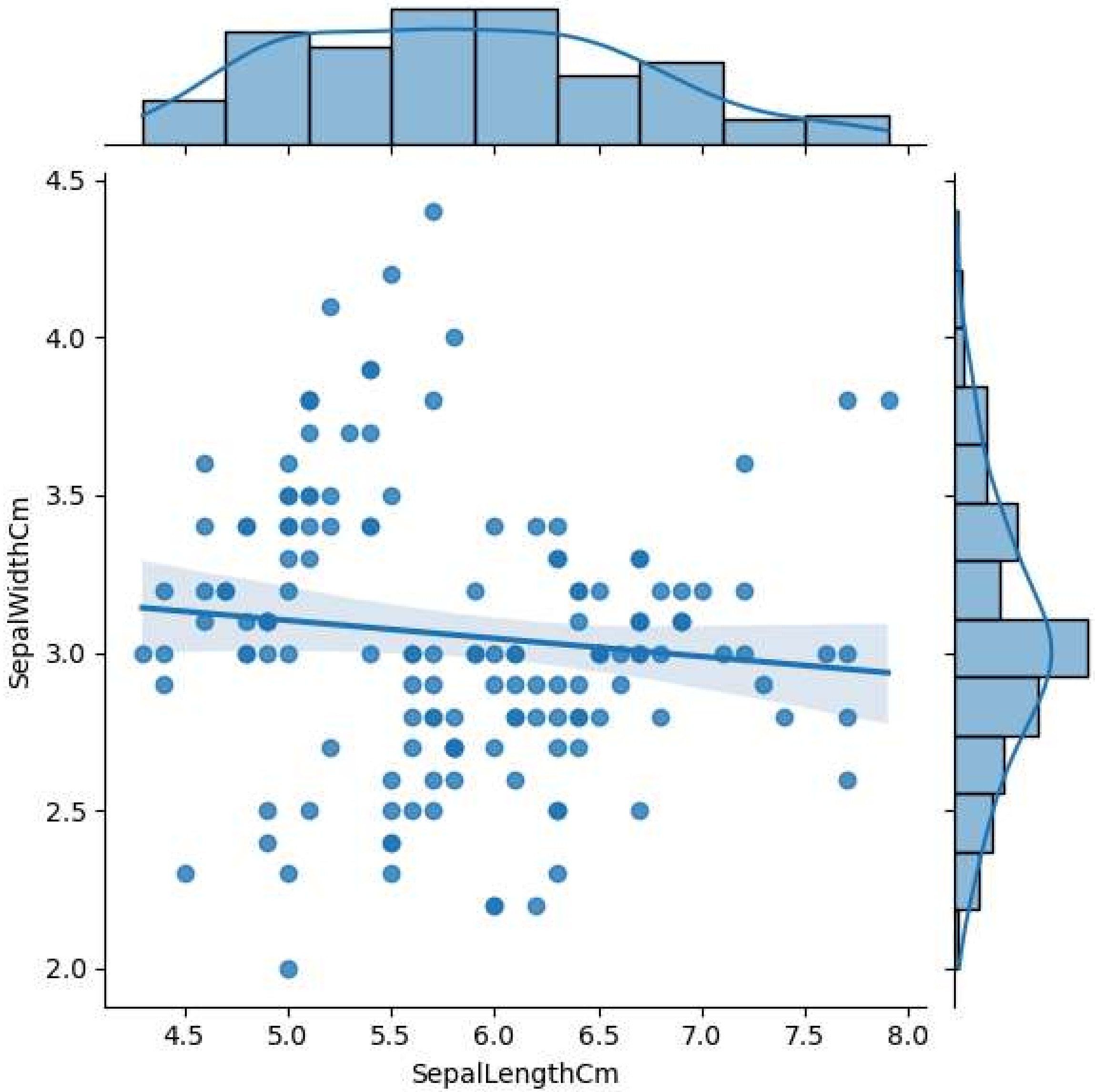
	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [22]:
```
fig = sns.jointplot(x = 'SepalLengthCm', y = 'SepalWidthCm', data = iris)
```



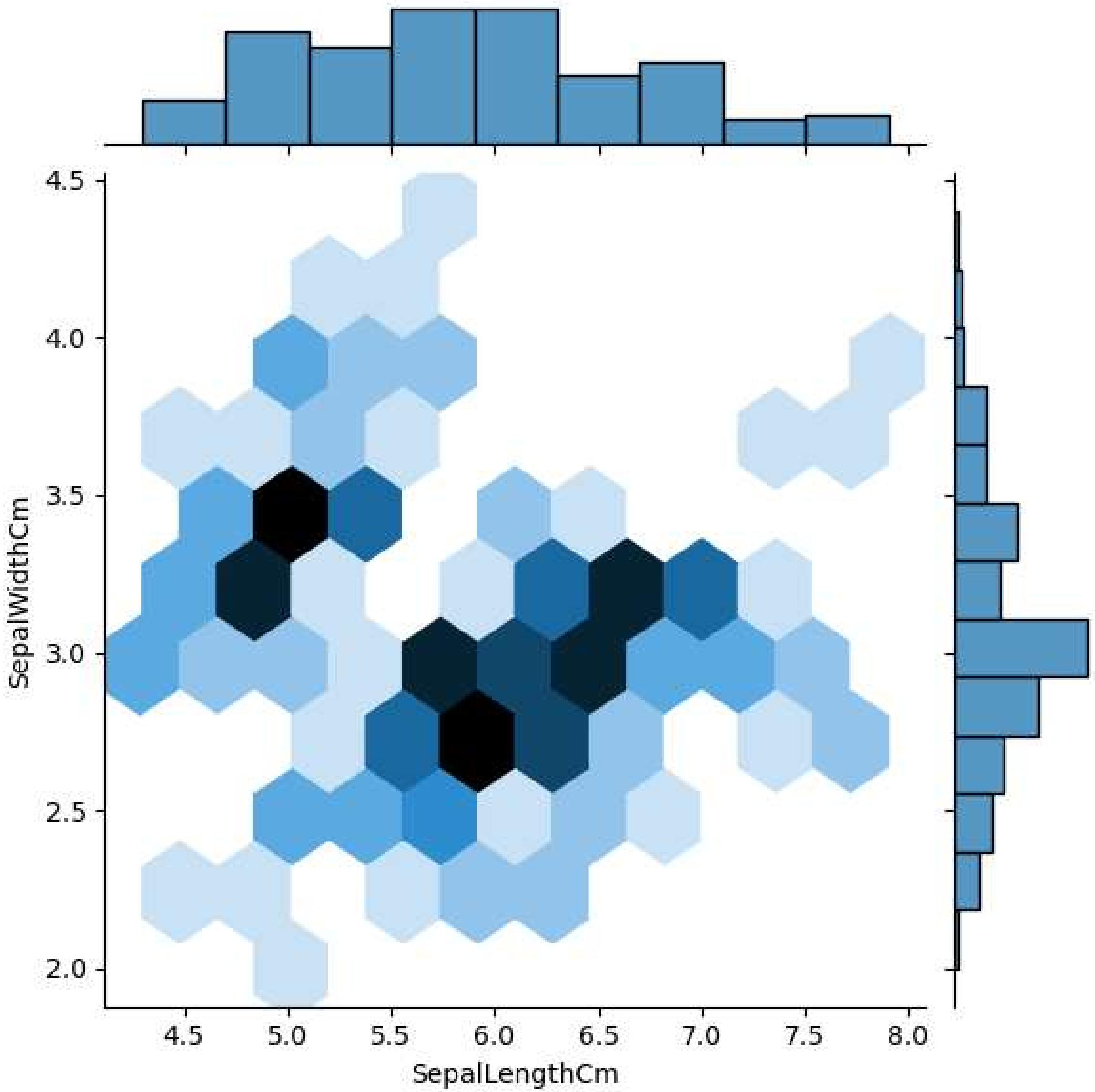
In [23]:
```
sns.jointplot(x = 'SepalLengthCm', y = 'SepalWidthCm', data = iris, kind = 'reg')
```

Out[23]:
```
<seaborn.axisgrid.JointGrid at 0x24206668690>
```



In [24]:
```
sns.jointplot(x = 'SepalLengthCm', y = 'SepalWidthCm', data = iris, kind = 'hex')
```

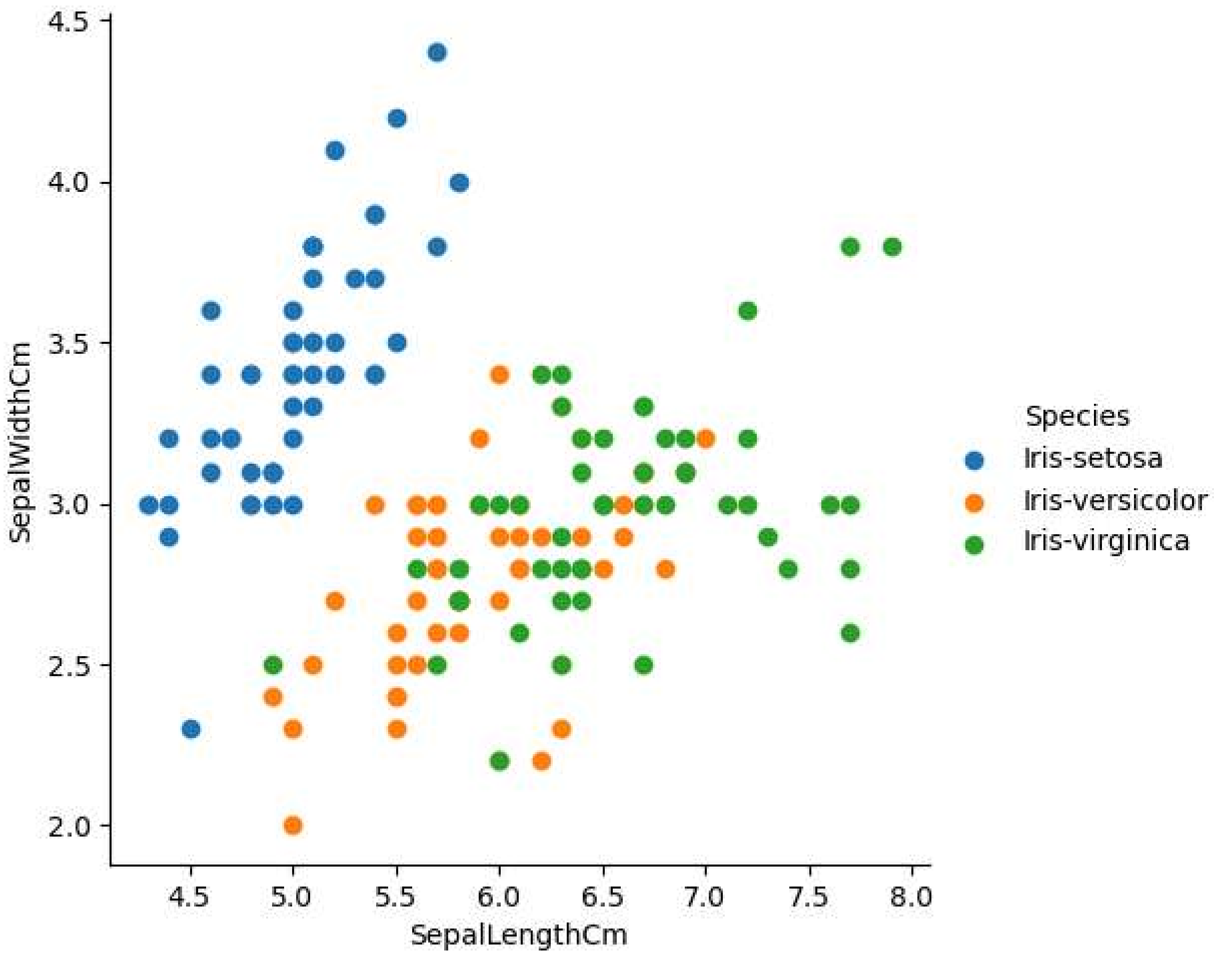
Out[24]: <seaborn.axisgrid.JointGrid at 0x2420670bed0>



In [36]:

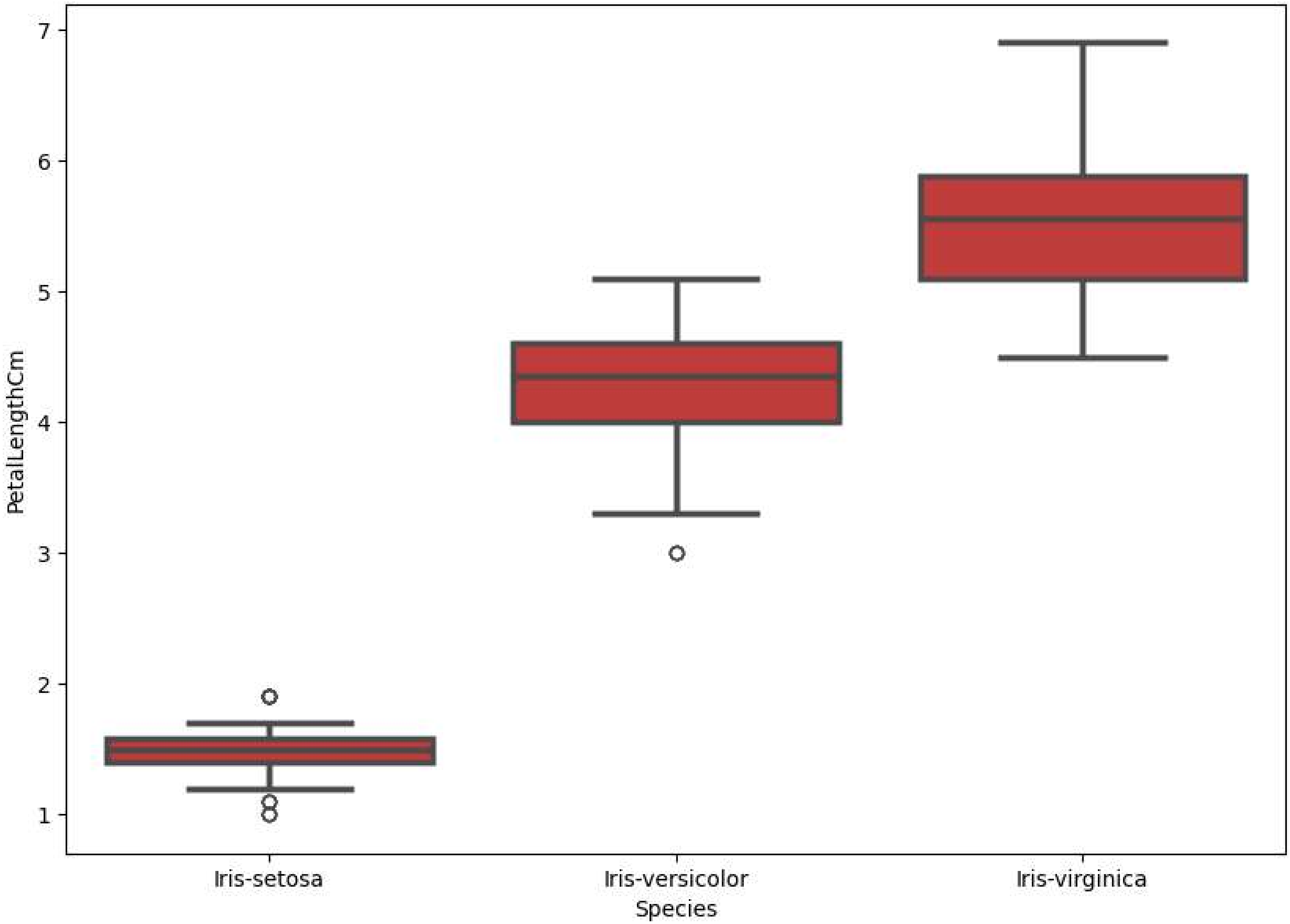
```
#Facet Grid
import matplotlib.pyplot as plt
%matplotlib inline

sns.FacetGrid(iris, hue='Species', height=5) \
    .map(plt.scatter, 'SepalLengthCm', 'SepalWidthCm') \
```

```
    .add_legend()
plt.show()
```

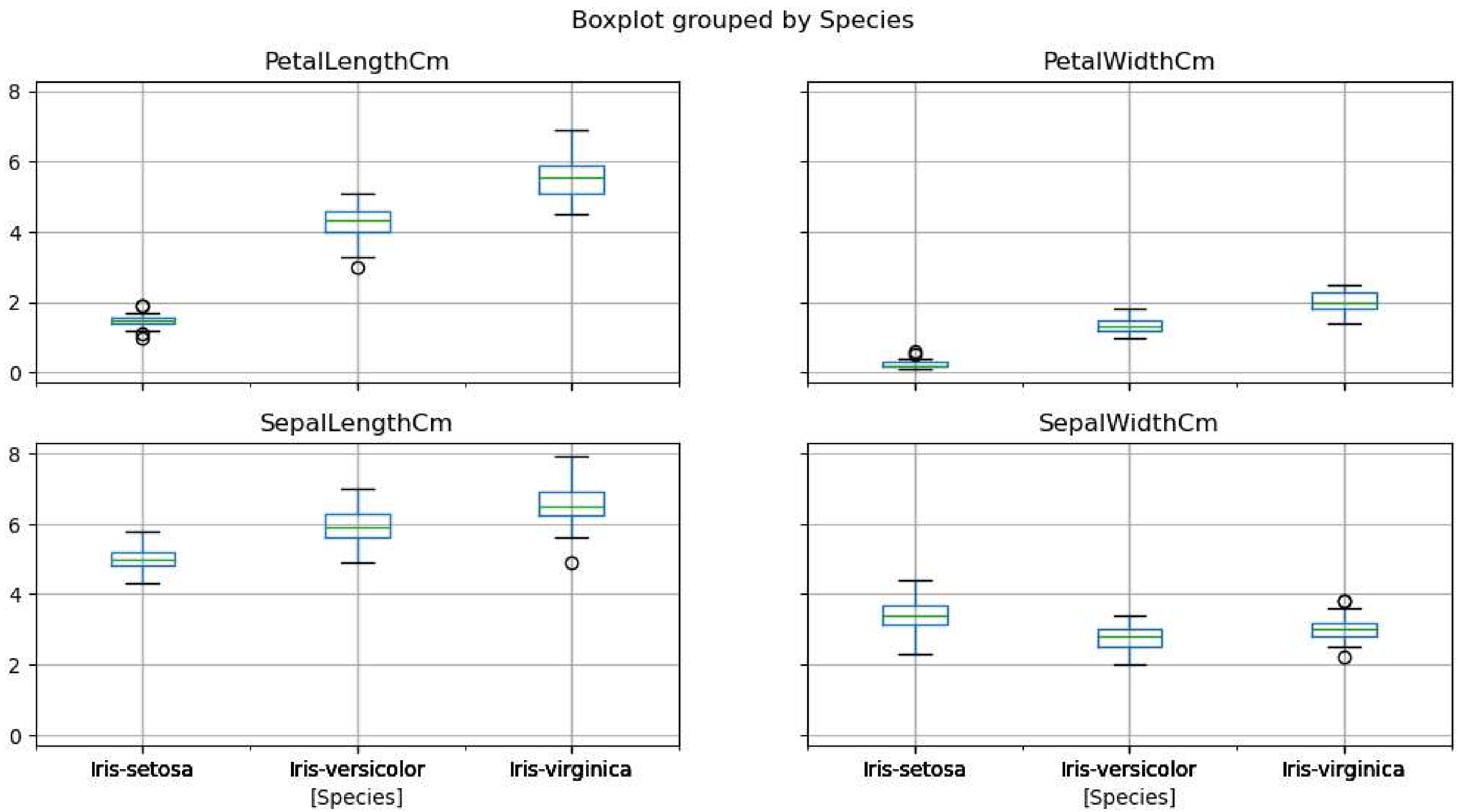


In [42]:

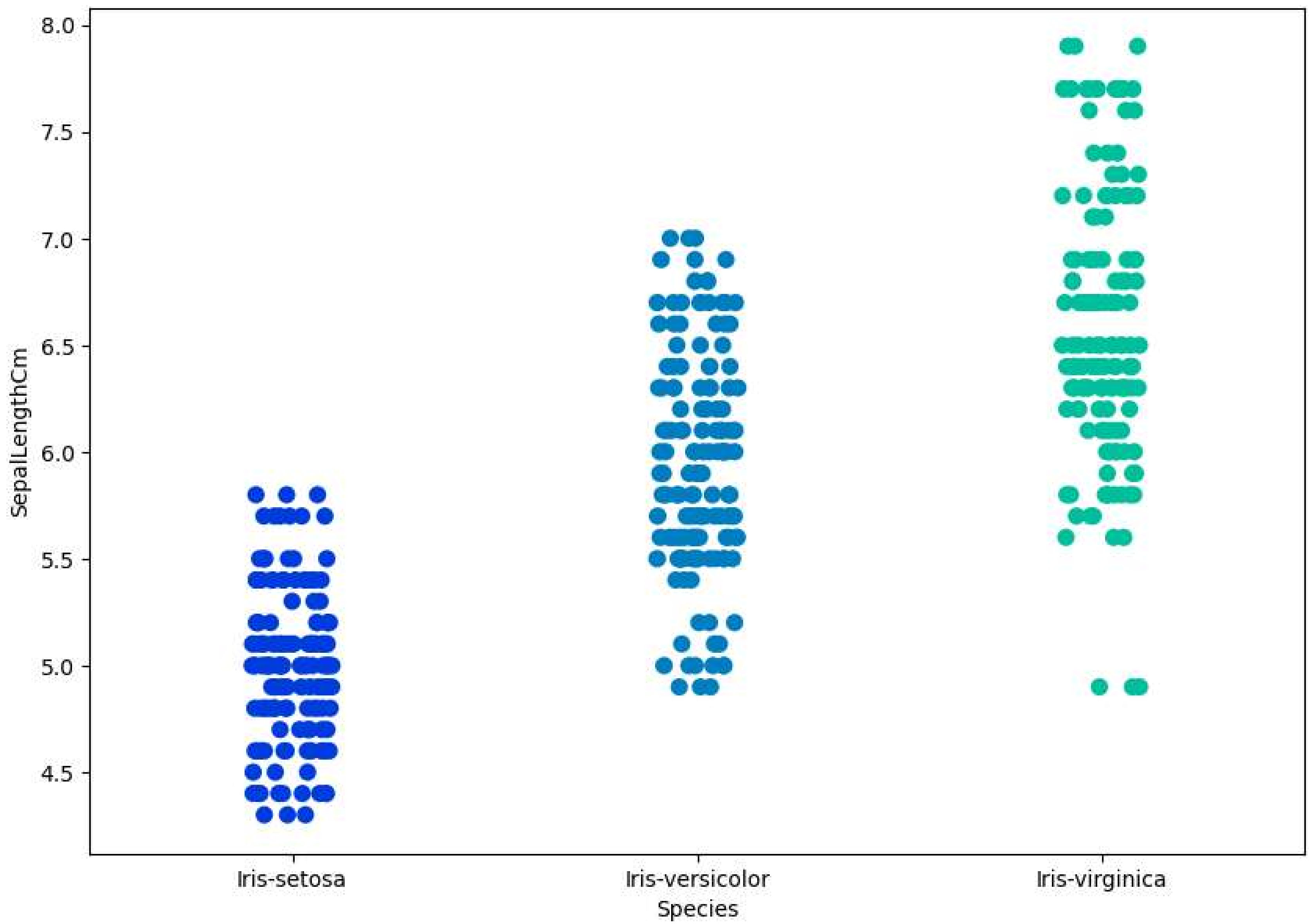
```
fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.boxplot(x = 'Species', y = 'PetalLengthCm', data = iris, order = ['Iris-virginica','Iris-versicolor','Iris-
plt.show()
```



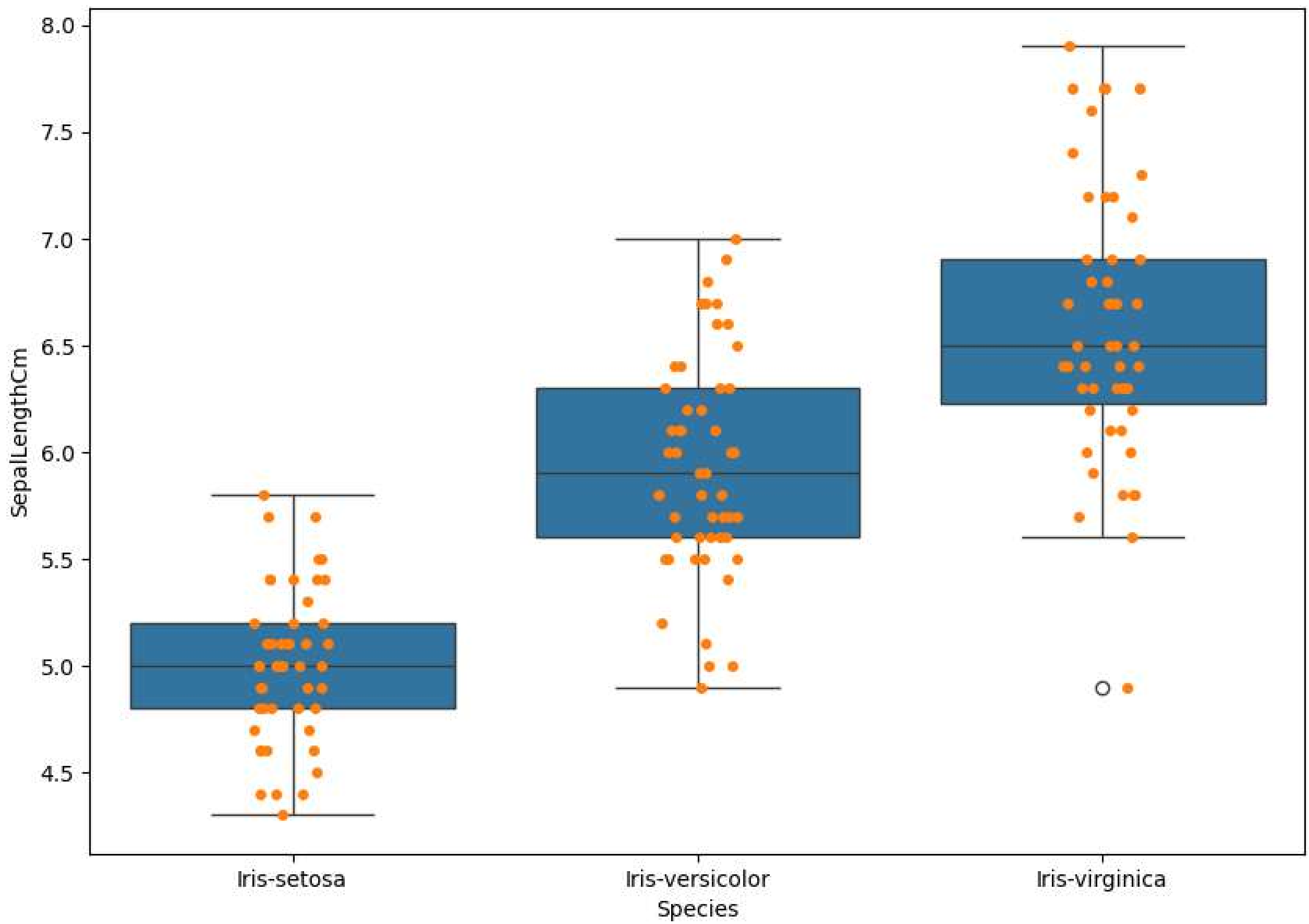
In [45]:
```
iris.boxplot(by = 'Species', figsize = (12,6))
plt.show()
```



In [49]:
```
fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.stripplot(x = 'Species', y = 'SepalLengthCm', data = iris, jitter = True, edgecolor = 'grey', size = 8, pal
plt.show()
```

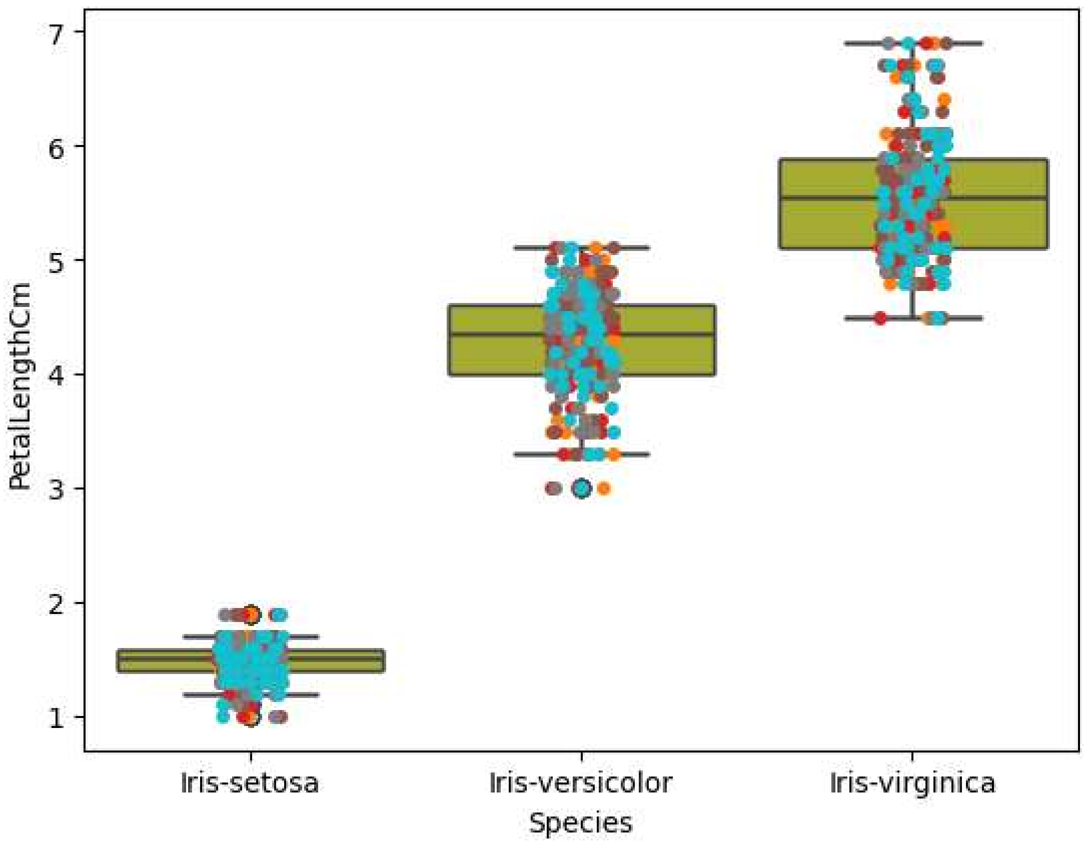


In [51]:
```
fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.boxplot(x = 'Species', y = 'SepalLengthCm', data = iris)
fig = sns.stripplot(x = 'Species', y = 'SepalLengthCm', data = iris, jitter = True, edgecolor = 'gray')
plt.show()
```

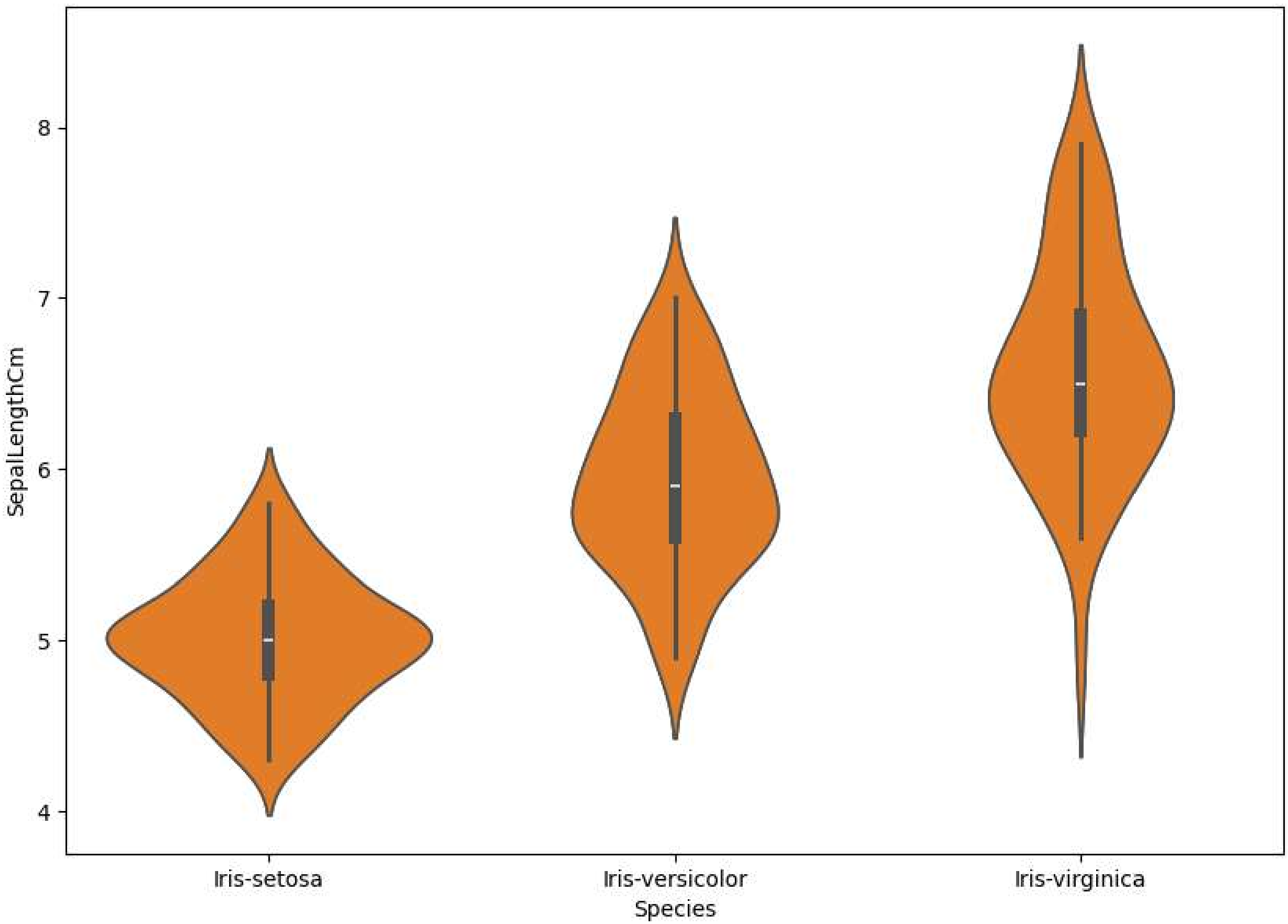


In [56]:
```
ax = sns.boxplot(x="Species", y="PetalLengthCm", data=iris)
sns.stripplot(x="Species", y="PetalLengthCm", data=iris, jitter=True, edgecolor="gray")

# The box patches (one per Species)
box1 = ax.patches[0]
```

```
box2 = ax.patches[1]
box3 = ax.patches[2]

# Set colors
box1.set_facecolor("green")
box1.set_edgecolor("black")

box2.set_facecolor("red")
box2.set_edgecolor("black")

box3.set_facecolor("yellow")
box3.set_edgecolor("black")

plt.show()
```

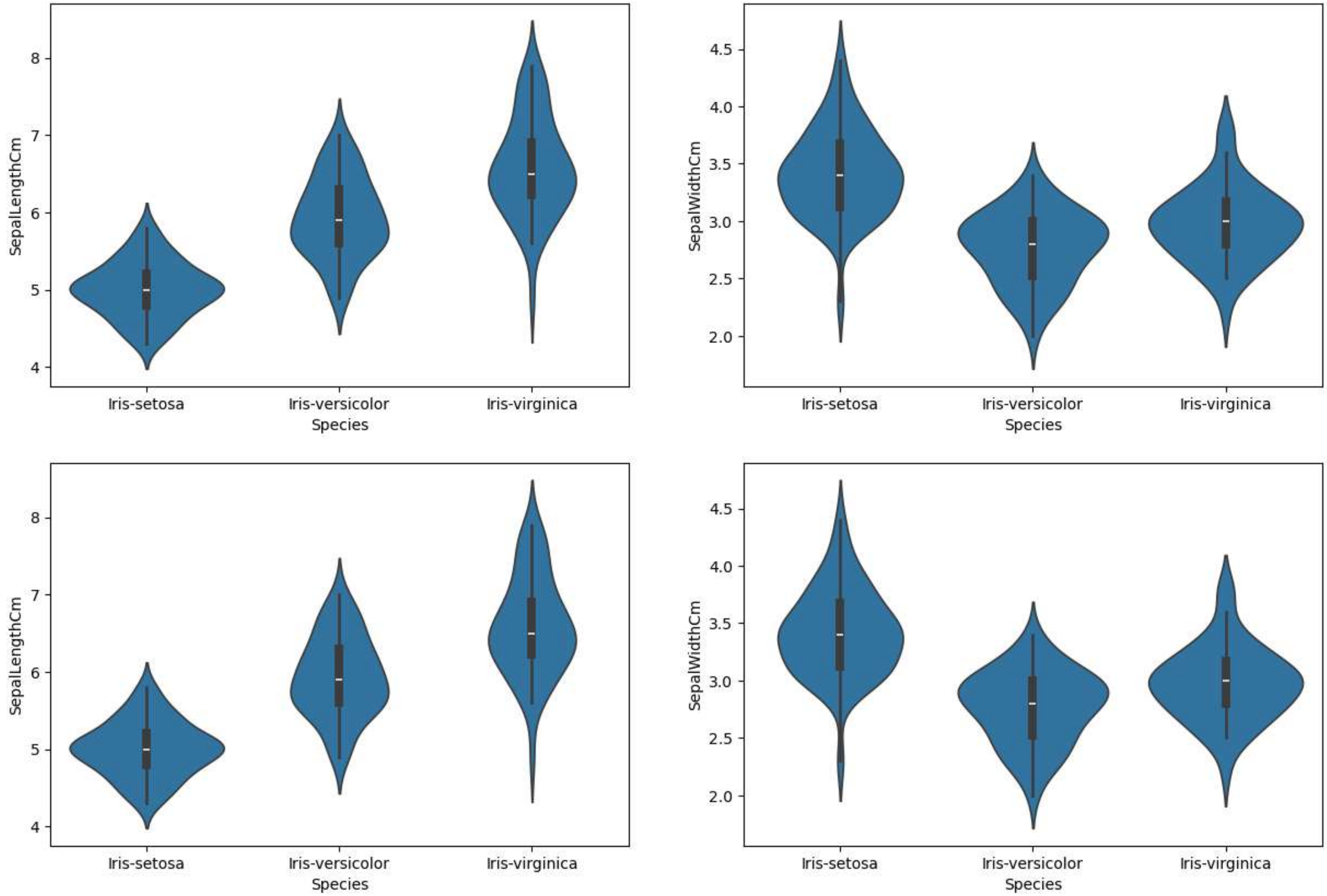


In [59]:
```
fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.violinplot(x = 'Species', y = 'SepalLengthCm', data = iris)
plt.show()
```

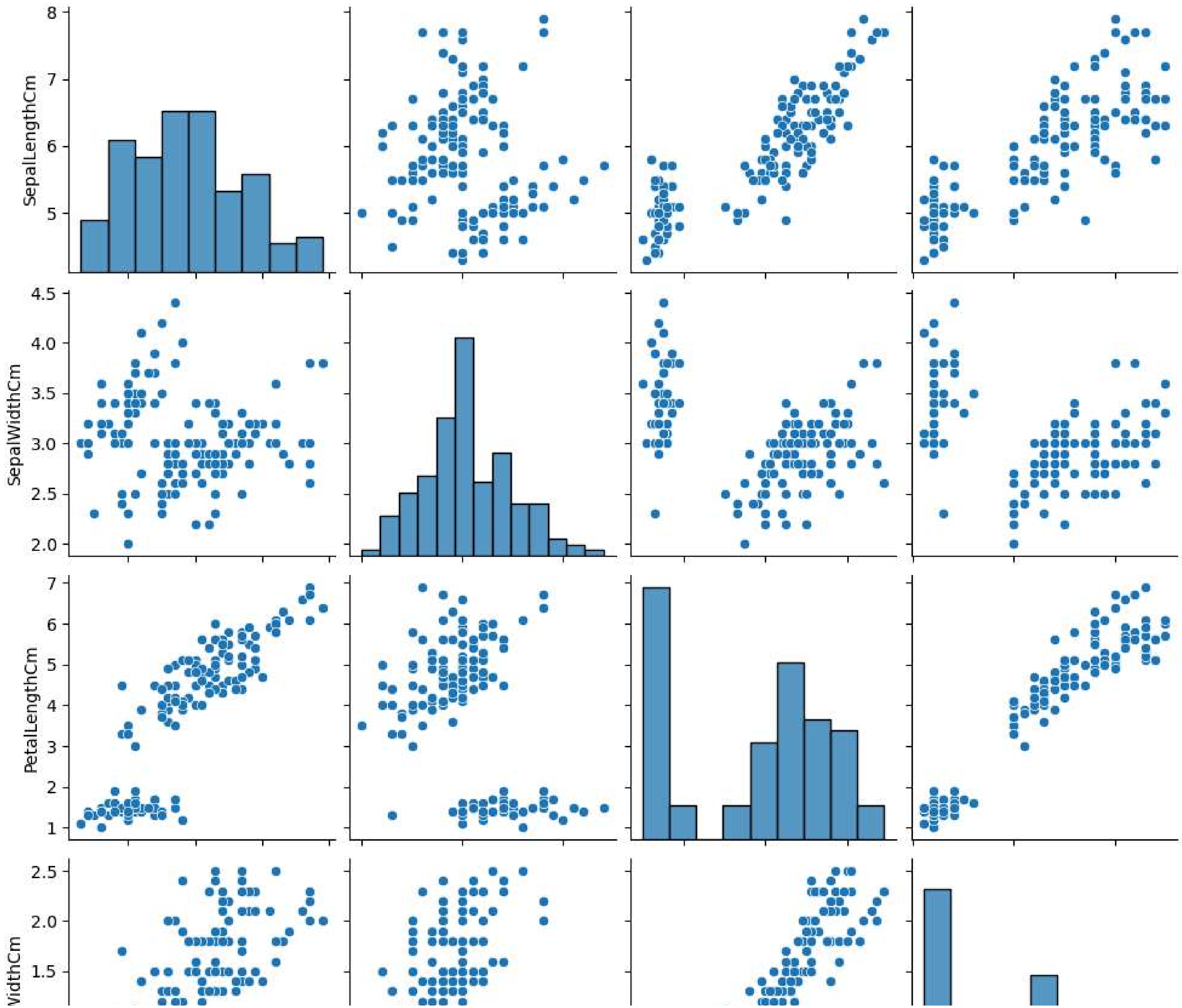


In [62]:

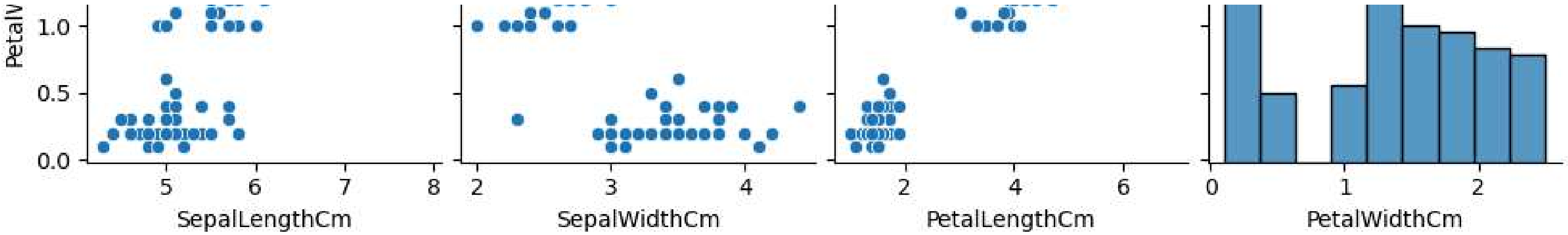
```
plt.figure(figsize = (15,10))
plt.subplot(2,2,1)
sns.violinplot(x = 'Species', y = 'SepalLengthCm', data = iris)
plt.subplot(2,2,2)
sns.violinplot(x = 'Species', y = 'SepalWidthCm', data = iris)
plt.subplot(2,2,3)
sns.violinplot(x = 'Species', y = 'SepalLengthCm', data = iris)
plt.subplot(2,2,4)
sns.violinplot(x = 'Species', y = 'SepalWidthCm', data = iris)

plt.show()
```



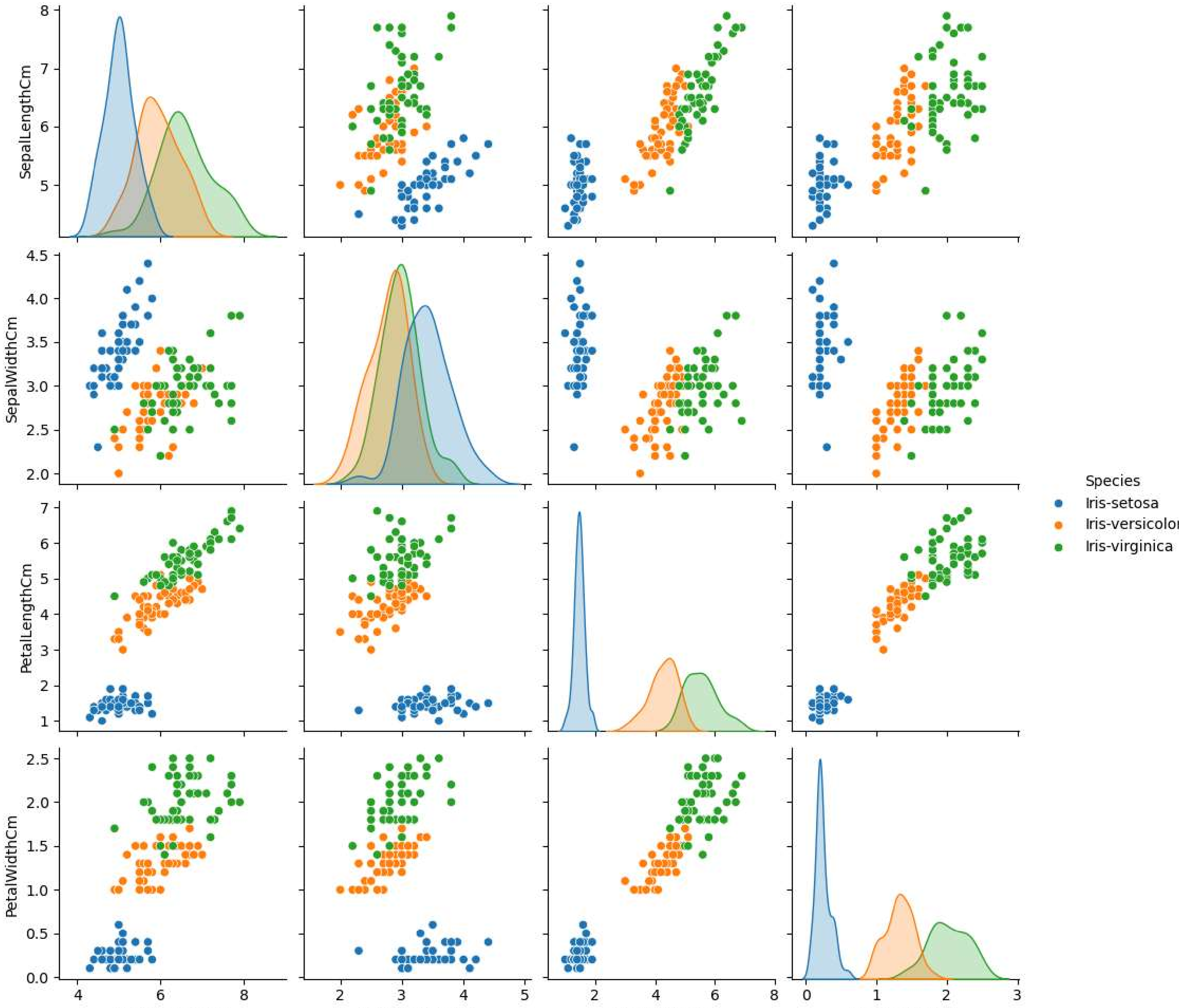
In [70]:
```
sns.pairplot(data = iris, kind = 'scatter')
plt.show()
```



SepalLengthCm
SepalWidthCm
PetalLengthCm
WidthCm
8
7
6
5
4.5
4.0
3.5
3.0
2.5
2.0
7
6
5
4
3
2
1
2.5
2.0
1.5

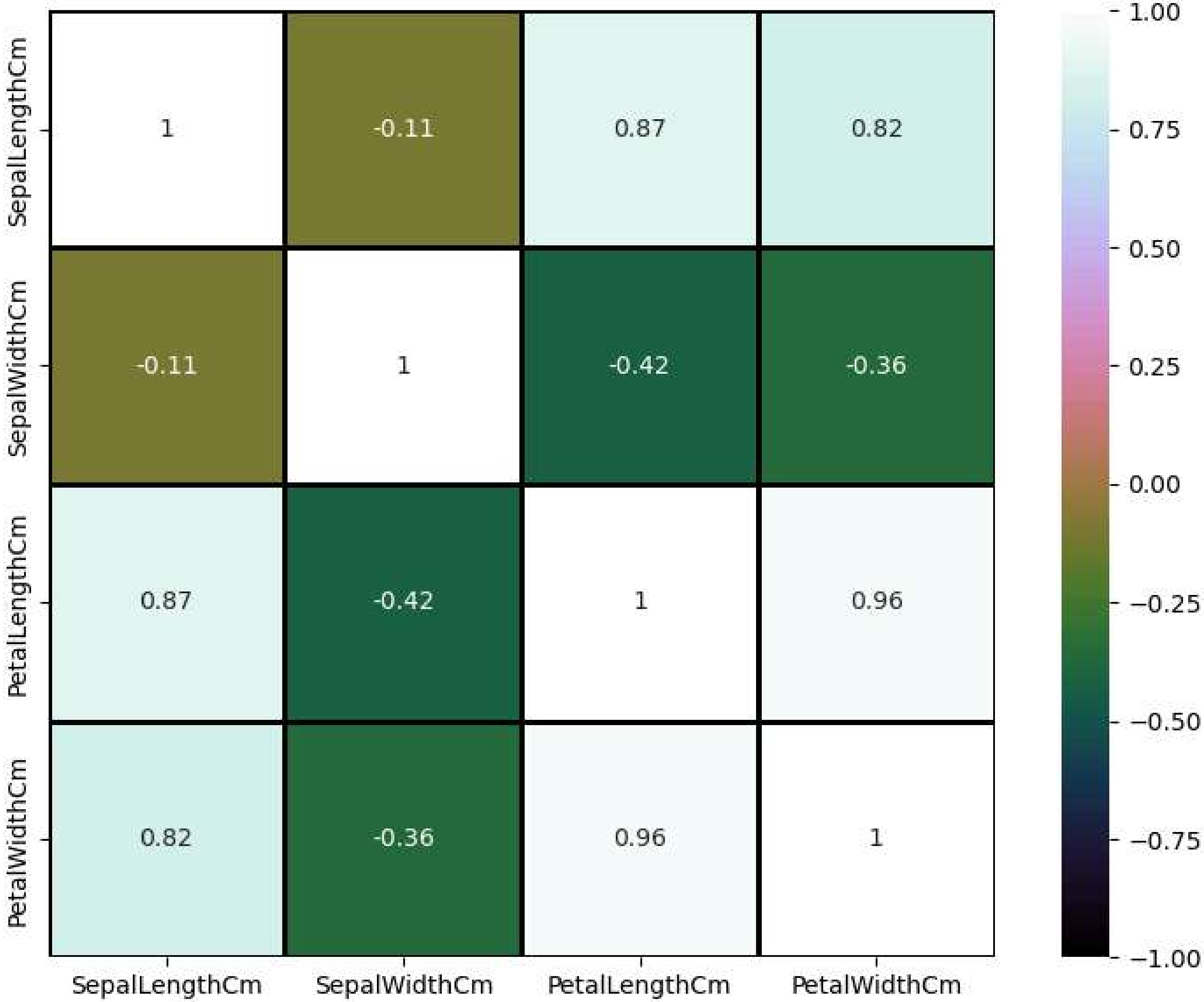


In [71]:
```
sns.pairplot(data = iris, hue = 'Species')
plt.show()
```

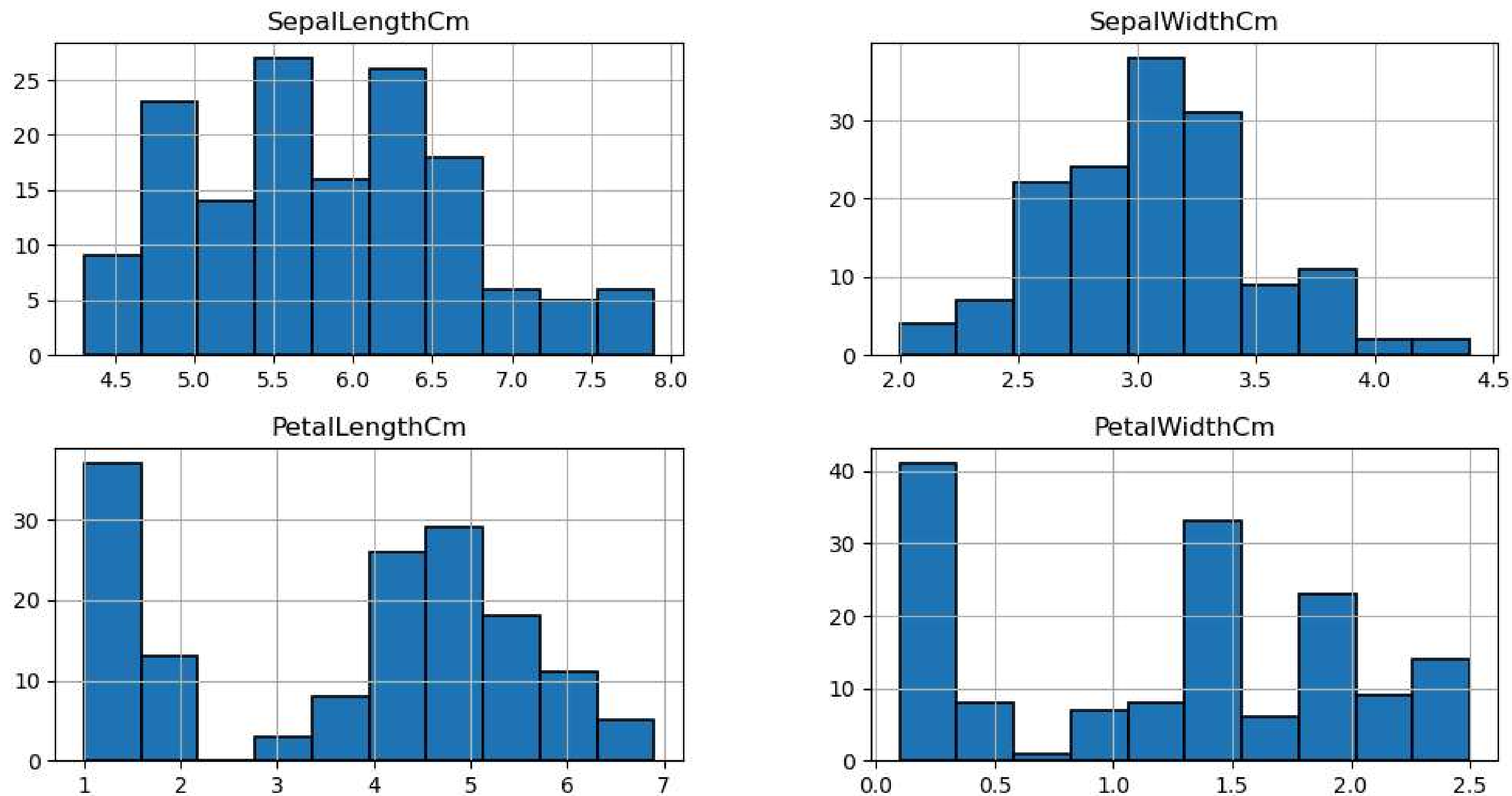

SepalLengthCm
SepalWidthCm
PetalLengthCm
PetalWidthCm
Species
Iris-setosa
Iris-versicolor
Iris-virginica

SepalLengthCm SepalWidthCm PetalLengthCm PetalWidthCm

In [77]:
```
fig = plt.gcf()
fig.set_size_inches(10, 7)

numeric_iris = iris.select_dtypes(include=['float64', 'int64'])

sns.heatmap(
    numeric_iris.corr(),
    annot=True,
    cmap="cubehelix",
    linewidths=1,
    linecolor='k',
    square=True,
    vmin=-1,
    vmax=1,
    cbar=True,
    cbar_kws={"orientation": "vertical"}
)

plt.show()
```

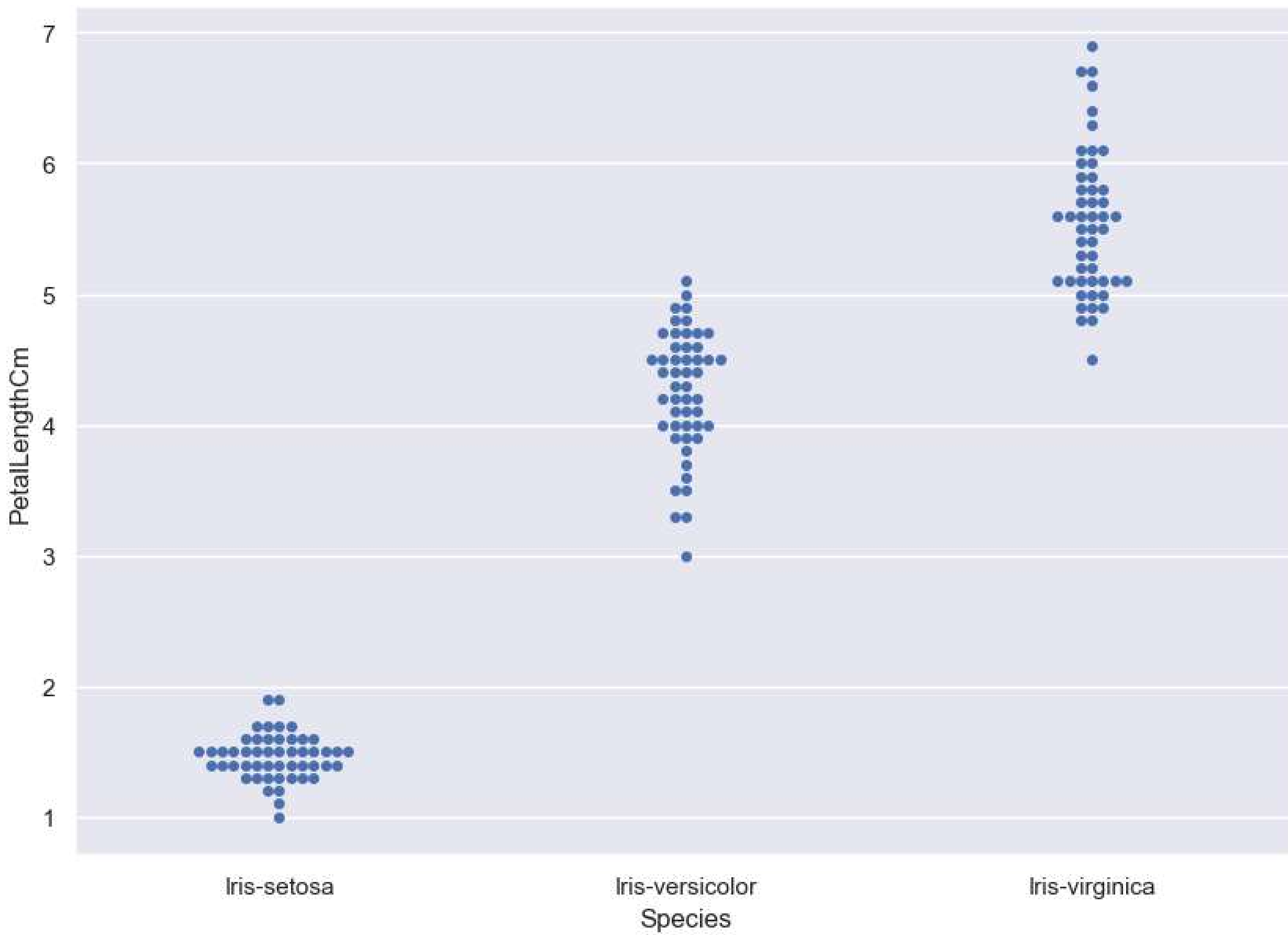


In [79]:
```
iris.hist(edgecolor = 'black', linewidth = 1.2)
fig = plt.gcf()
fig.set_size_inches(12,6)
plt.show()
```



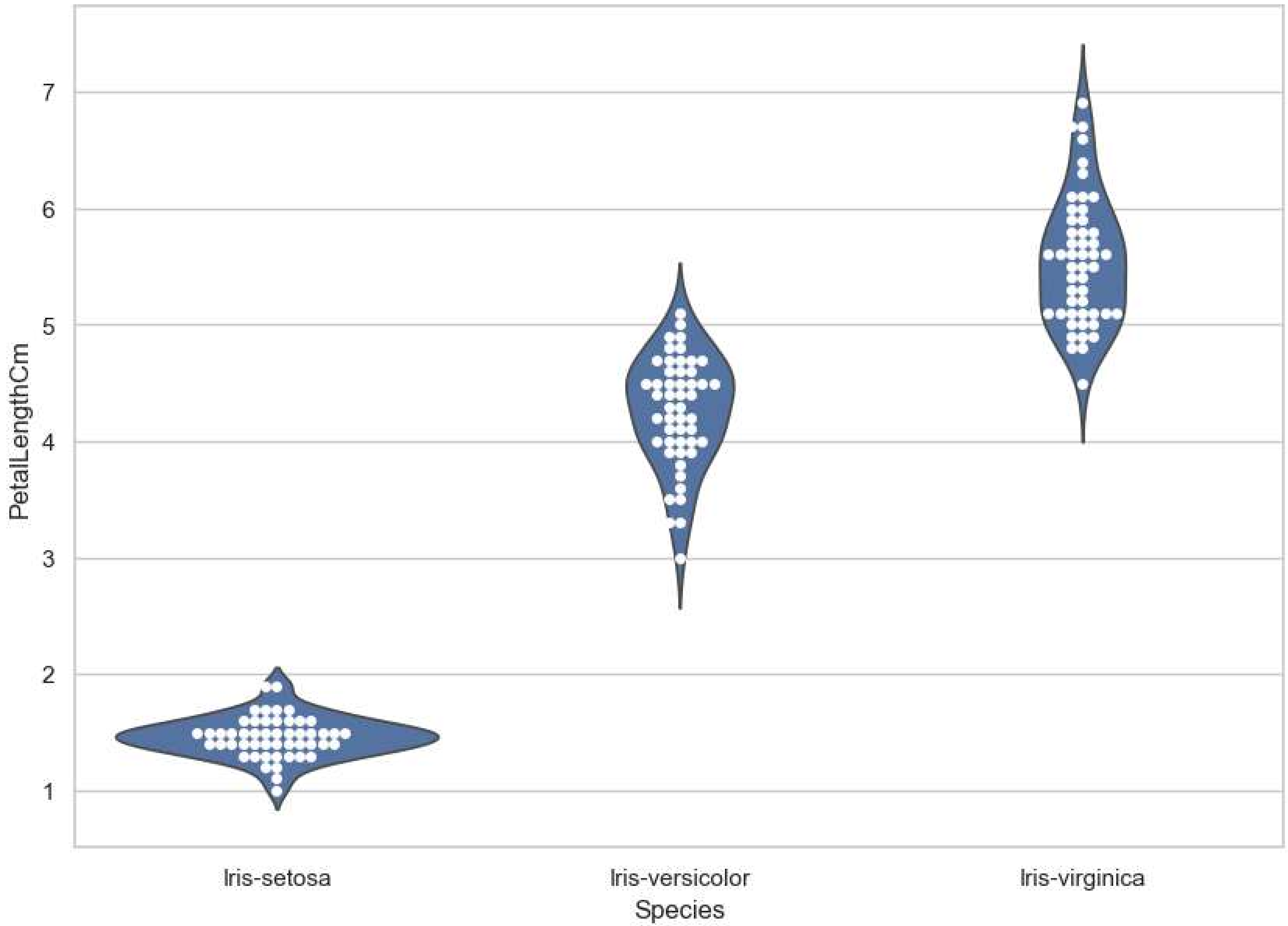
In [81]:

```
sns.set(style="darkgrid")
fig = plt.gcf()
fig.set_size_inches(10,7)
fig = sns.swarmplot(x = "Species", y = "PetalLengthCm", data = iris)
plt.show()
```

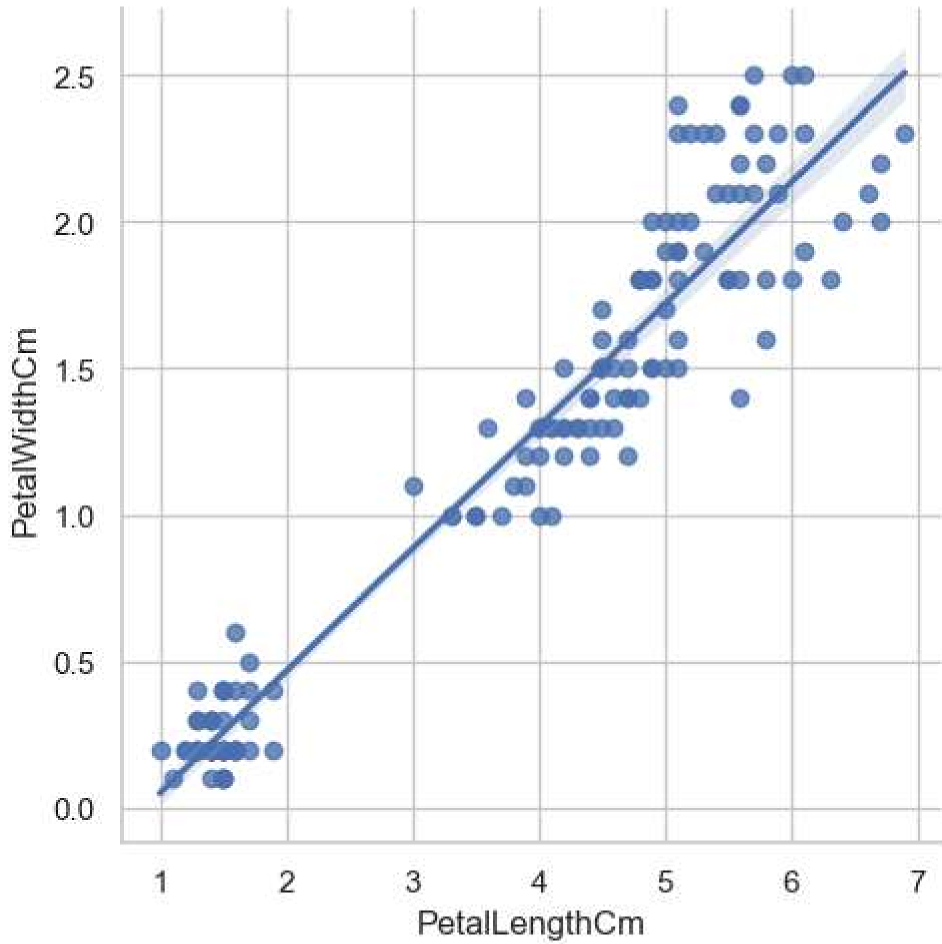


In [82]:
```
sns.set(style="whitegrid")
fig = plt.gcf()
fig.set_size_inches(10,7)
ax = sns.violinplot(x = "Species", y = "PetalLengthCm", data = iris, inner = None)
```

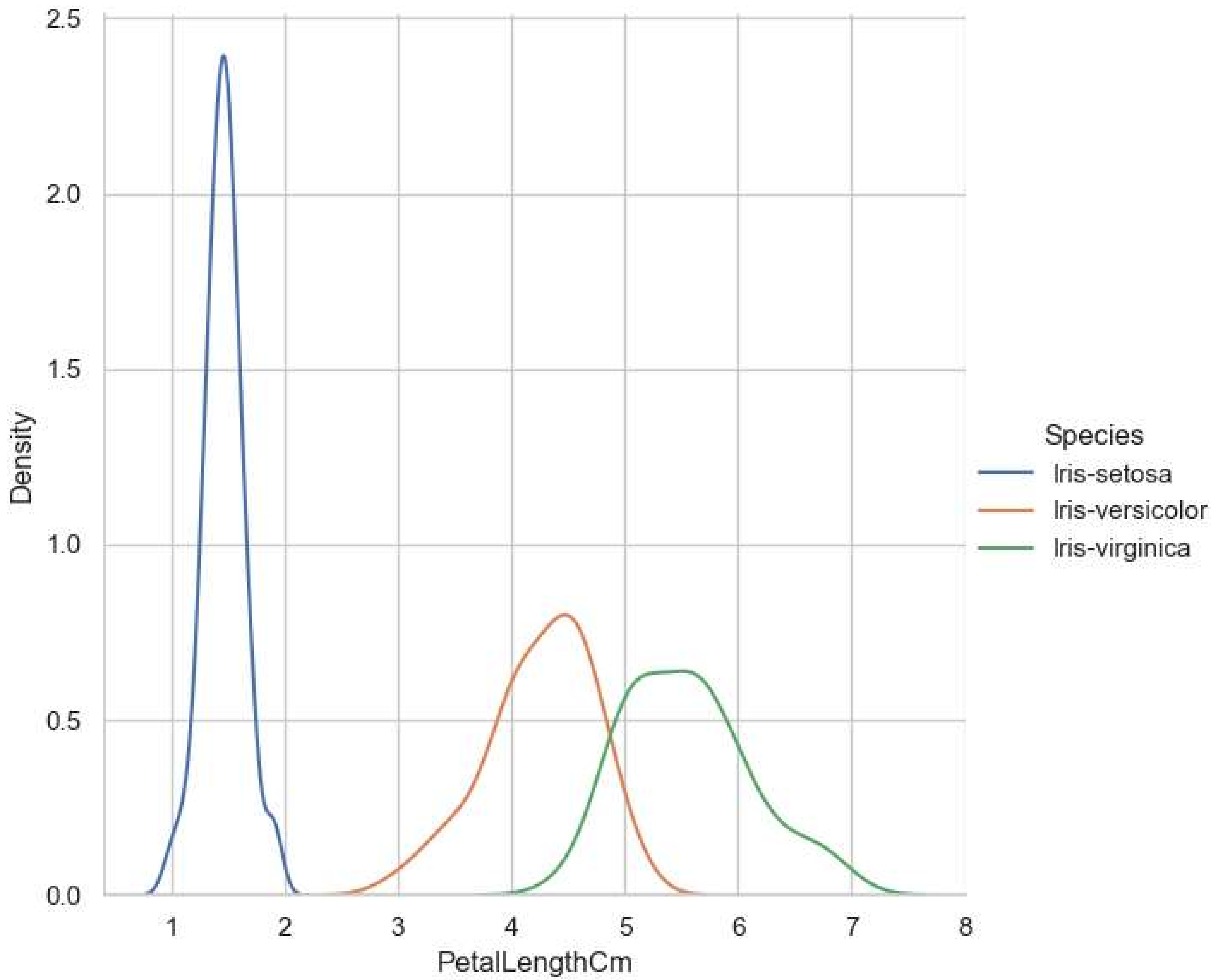
```
ax = sns.swarmplot(x = "Species", y = "PetalLengthCm", data = iris, color = "white", edgecolor = "black")
plt.show()
```



In [83]:
```
fig = sns.lmplot(x = "PetalLengthCm", y = "PetalWidthCm", data = iris)
plt.show()
```



In [92]:
```
#Facet Grid
sns.FacetGrid(iris, hue = "Species", height = 6)\
    .map(sns.kdeplot, "PetalLengthCm")\
    .add_legend()
plt.ioff()
plt.show()
```

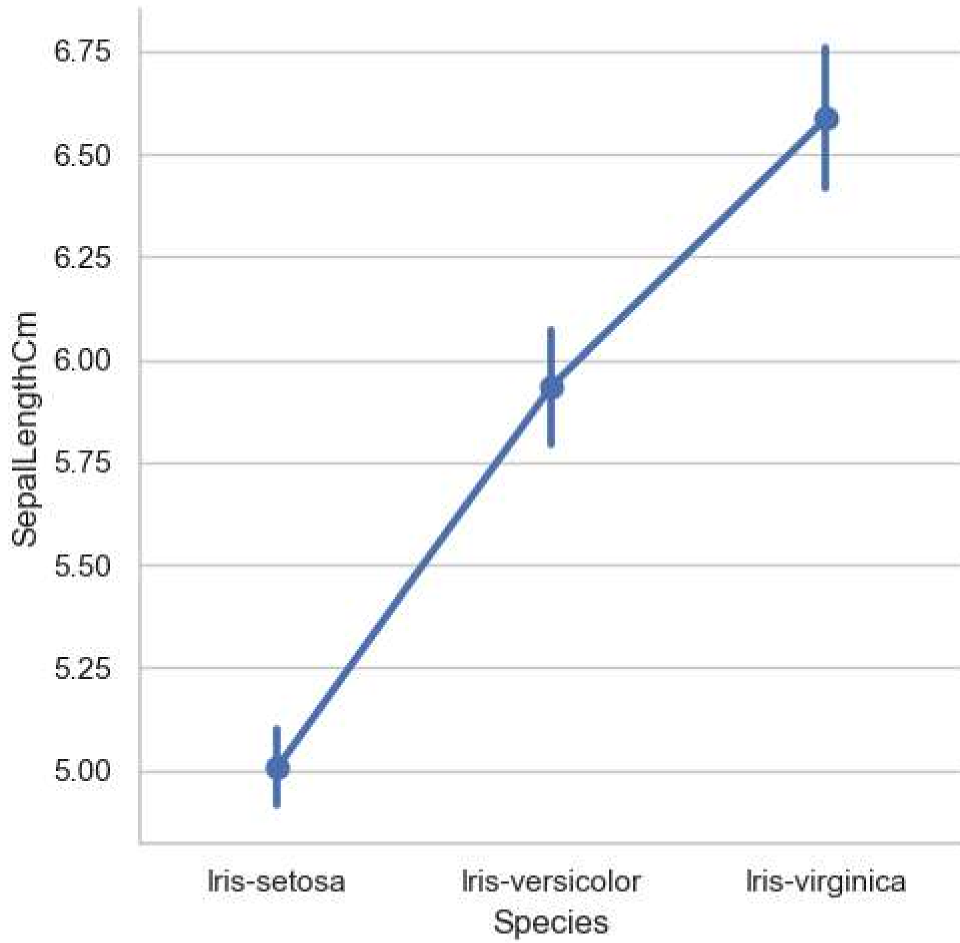


In [95]:
```
#fig = sns.factorplot(x = 'Species', y = 'SepalLengthCm', data = iris)
#plt.ioff()
#plt.show()
```

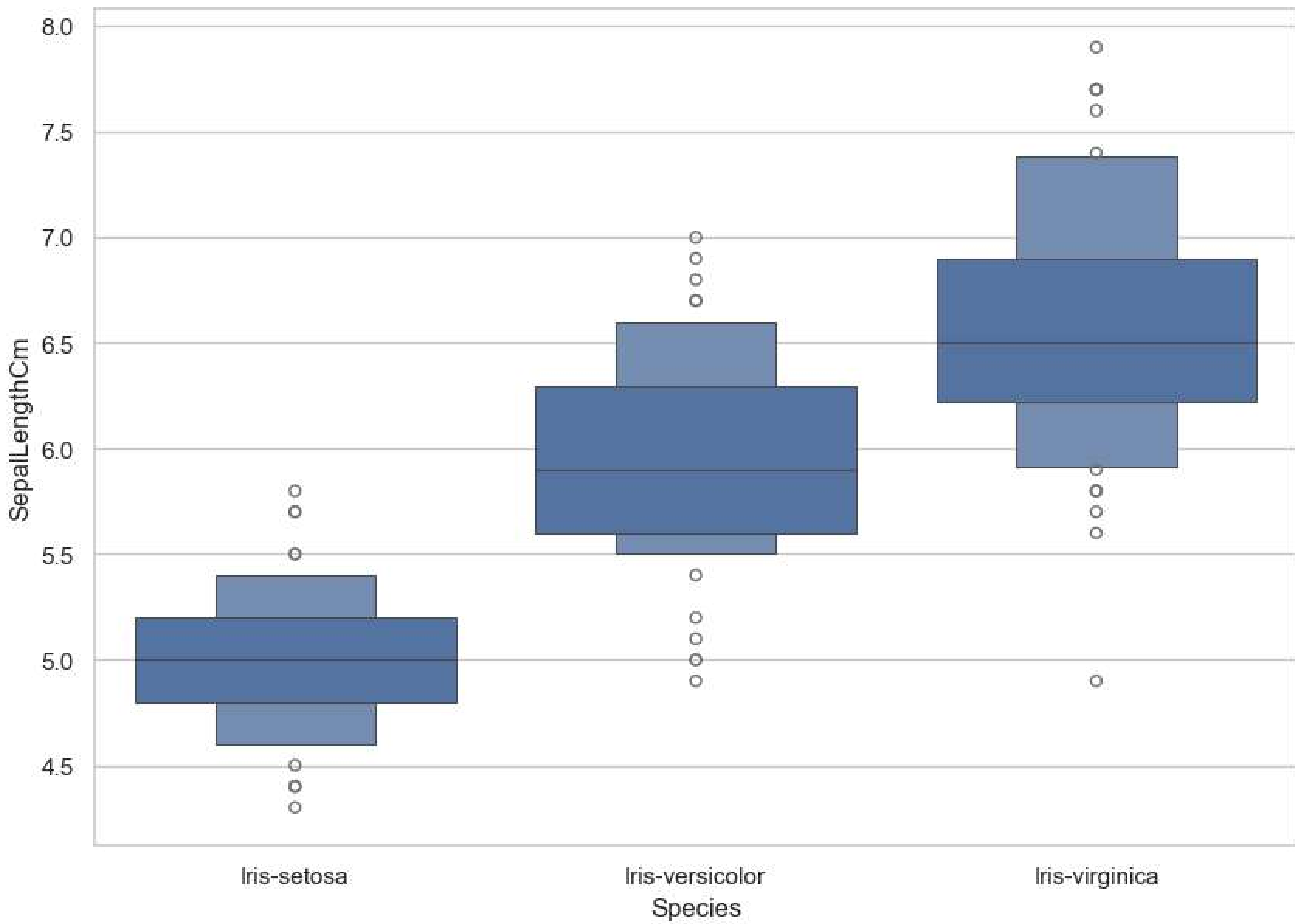
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[95], line 1
----> 1 fig = sns.factorplot(x = 'Species', y = 'SepalLengthCm', data = iris)
      2 #plt.ioff()
      3 plt.show()

AttributeError: module 'seaborn' has no attribute 'factorplot'
```

In [97]:

```
#factorplot() replaced with catplot()
fig = sns.catplot(x = 'Species', y = 'SepalLengthCm', data = iris, kind = 'point')
plt.show()
```

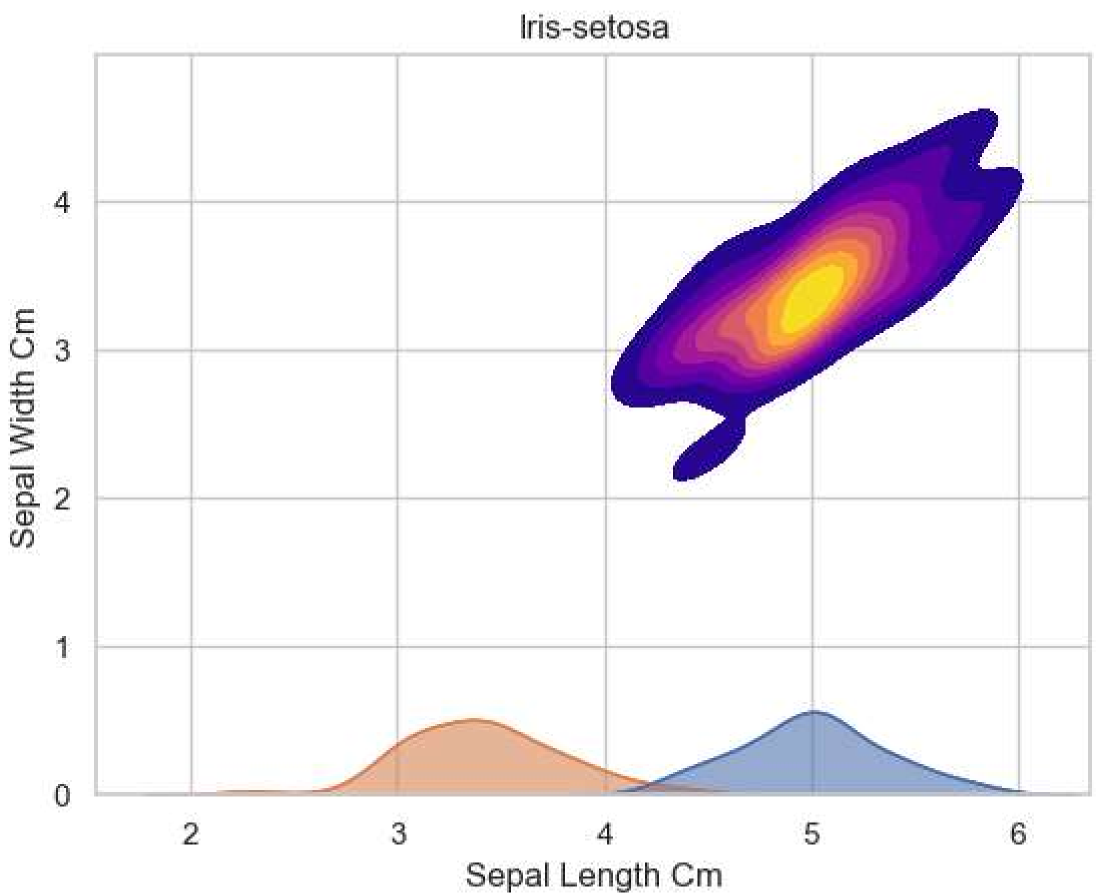


In [98]:
```python
fig=plt.gcf()
fig.set_size_inches(10,7)
fig=sns.boxenplot(x = 'Species', y = 'SepalLengthCm', data = iris)
plt.show()
```



In [104…

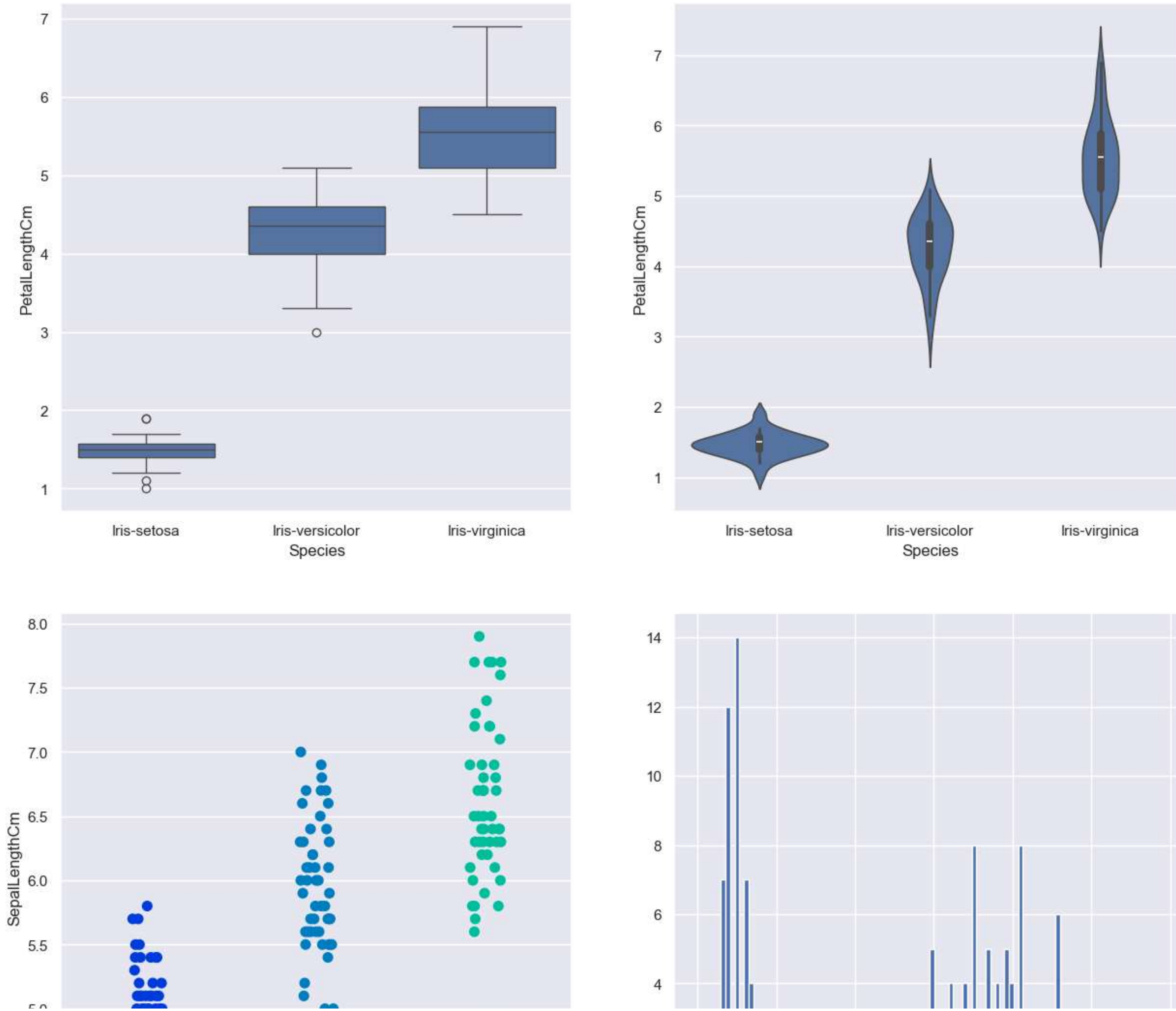
```
sub = iris[iris['Species'] == 'Iris-setosa']
sns.kdeplot(data = sub, x = 'SepalLengthCm', y = 'SepalWidthCm',cmap = "plasma", fill = True, thresh = 0.05)
plt.title('Iris-setosa')
plt.xlabel('Sepal Length Cm')
plt.ylabel('Sepal Width Cm')
plt.show()
```

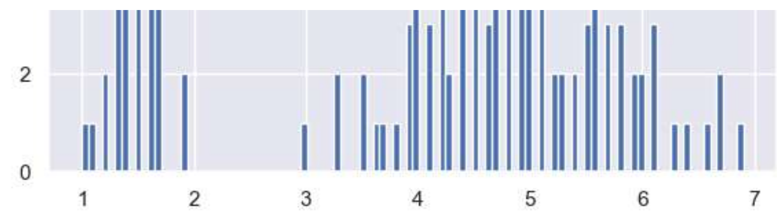


In [105…

```
sns.set_style('darkgrid')
f,axes = plt.subplots(2,2,figsize=(15,15))

k1 = sns.boxplot(x = "Species", y = "PetalLengthCm", data = iris, ax = axes[0,0])
k2 = sns.violinplot(x = 'Species', y = 'PetalLengthCm', data = iris, ax = axes[0,1])
k3 = sns.stripplot(x = 'Species', y = 'SepalLengthCm', data = iris, jitter = True, edgecolor = 'gray', size = 8, pale
```

```
axes[1,1].hist(iris.PetalLengthCm,bins = 100)

plt.show()
```


PetalLengthCm
Iris-setosa
Iris-versicolor
Iris-virginica
Species
PetalLengthCm
Iris-setosa
Iris-versicolor
Iris-virginica
Species
SepalLengthCm

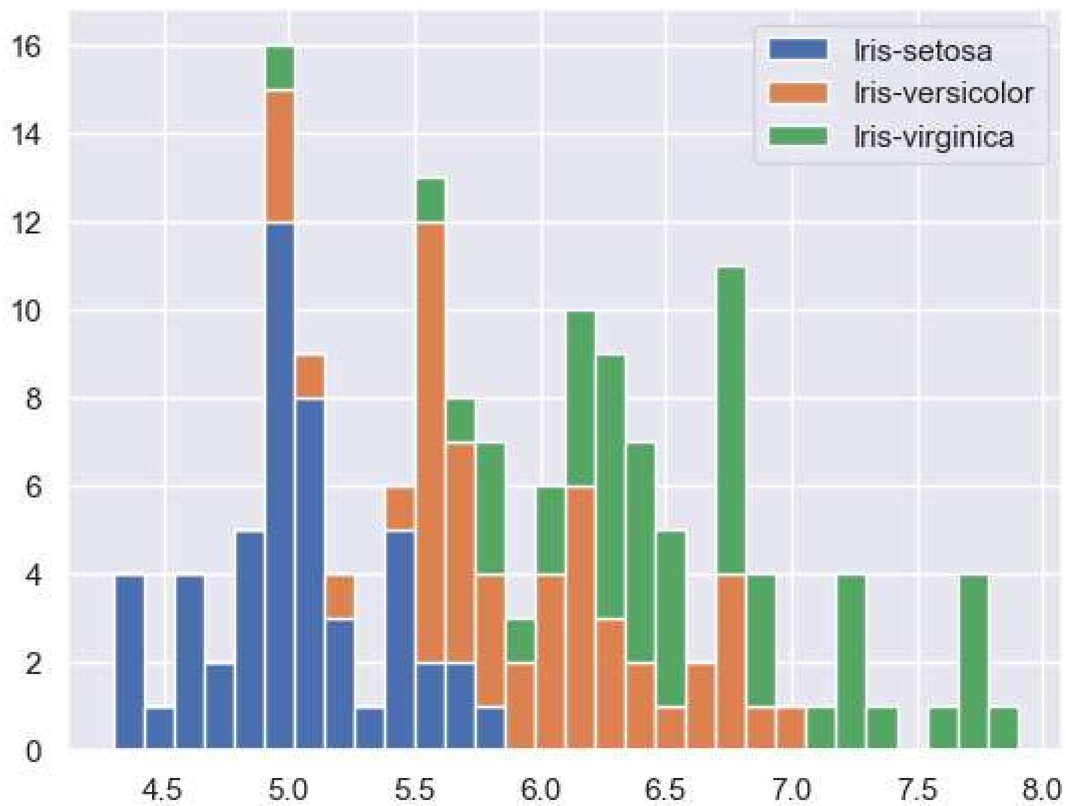


In [106…

```
iris['Species'] = iris['Species'].astype('category')
```

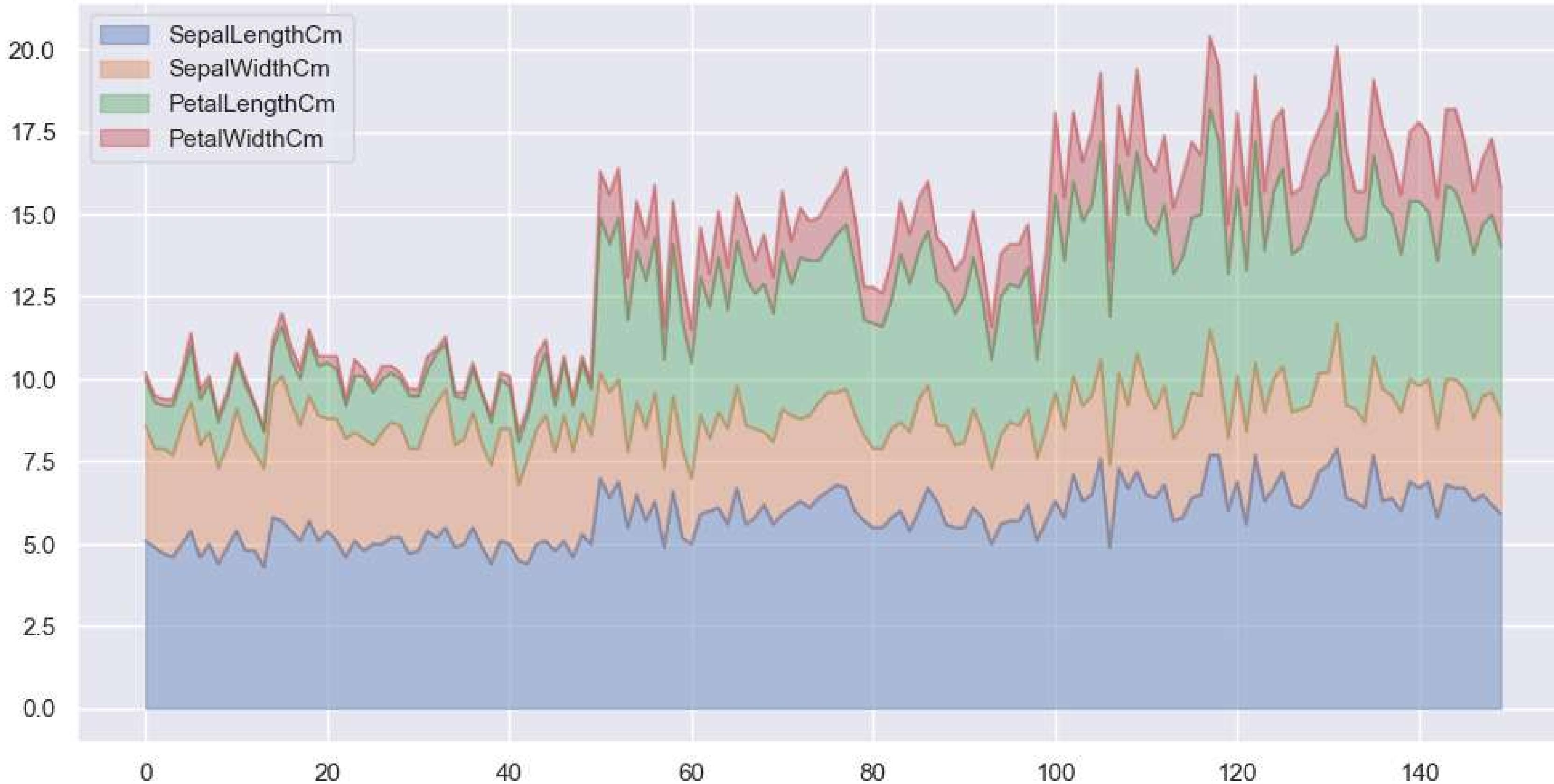
In [107…

```
list1 = list()
mylabels = list()
for gen in iris.Species.cat.categories:
    list1.append(iris[iris.Species == gen].SepalLengthCm)
    mylabels.append(gen)

h = plt.hist(list1, bins = 30, stacked = True, rwidth = 1,label = mylabels)
plt.legend()
plt.show()
```



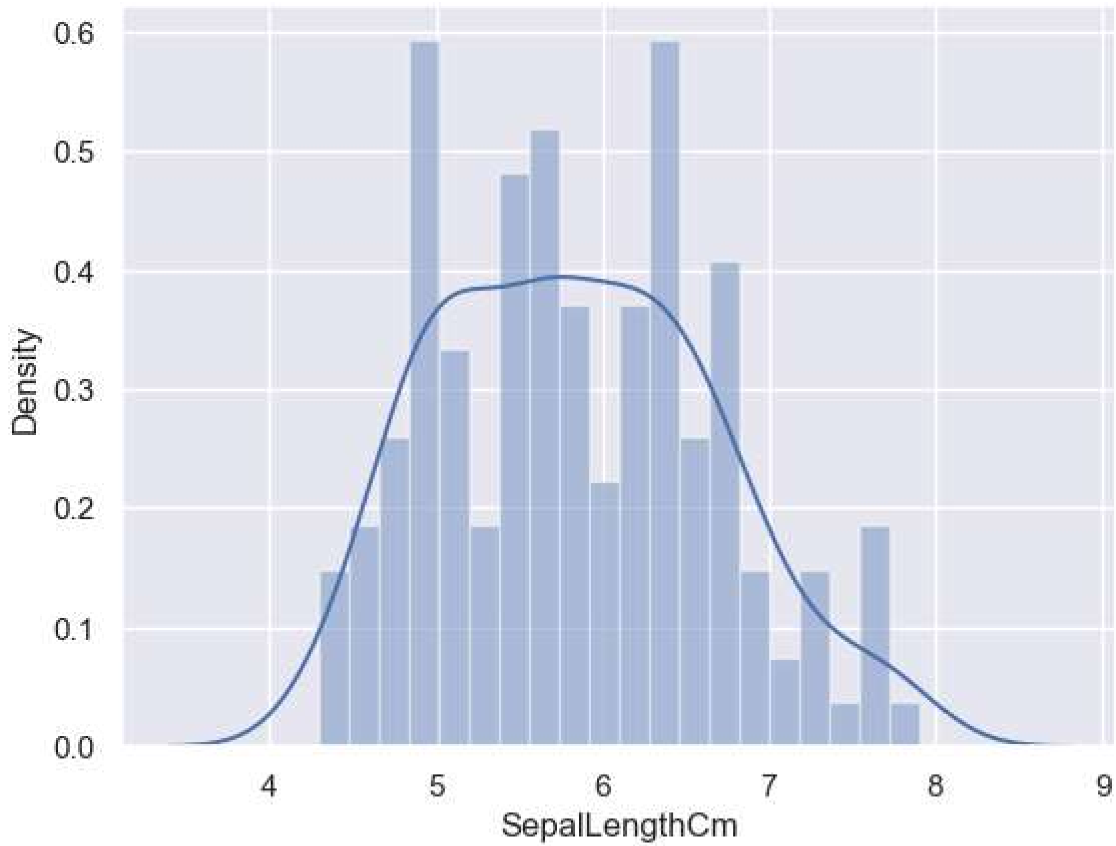
In [108…

```
iris.plot.area(y = ['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm'], alpha = 0.4, figsize = (12, 6))
plt.show()
```



In [109…

```
sns.distplot(iris['SepalLengthCm'], kde = True, bins = 20)
plt.show()
```




```
In [ ]:
```